

ArbitrageAI Platform — Non-Functional Requirements Document (NFRD)

Classification: Internal — Engineering
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Prepared for: Technical Lead & Engineering Team

1. Performance Requirements

1.1 Latency

ID	Requirement	Target	Measurement Method
PERF-01	Trade signal-to-order latency	p99 < 500ms	Cloud Monitoring custom metric: signal_emit_ts → order_submit_ts
PERF-02	Spread detection latency	< 50ms from WebSocket price update to signal emission	Internal trace via Cloud Trace
PERF-03	Vertex AI inference latency	< 50ms per prediction (online endpoint)	Vertex AI prediction latency metric
PERF-04	Dashboard page load	p95 < 2s (web), p95 < 1.5s (mobile)	Web Vitals (LCP); Firebase Performance SDK
PERF-05	P&L update to dashboard	< 3s from trade close to dashboard refresh	Firestore listener propagation time
PERF-06	Exchange connection validation	< 2s from API key paste to confirmation	Backend API response time
PERF-07	Notification delivery — WhatsApp	< 5s from trade close to WhatsApp message received	Cloud Run dispatcher latency
PERF-08	Notification delivery — SMS	< 15s from trigger to SMS received	Twilio delivery timestamp
PERF-09	Subscription activation	< 60s from payment confirmation to tier active	Webhook → Firestore update latency
PERF-10	Exchange reconnection	< 30s after WebSocket disconnect	CCXT Pro reconnect loop timing

1.2 Throughput

ID	Requirement	Target
THRU-01	Concurrent active strategy workers	1,000,000
THRU-02	Strategy evaluations per second	500,000

ID	Requirement	Target
THRU-03	Trade records written per day	50,000,000
THRU-04	Exchange API calls per minute	200,000 (across all users)
THRU-05	Concurrent dashboard WebSocket sessions	1,000,000
THRU-06	WhatsApp notifications per second	10,000 (burst)
THRU-07	Firestore reads per second	1,000,000

2. Availability & Reliability

2.1 Uptime SLOs

ID	Service	Target SLO	Measurement Window
AVAIL-01	Control plane (auth, dashboard, config) — Starter/Pro	99.9%	Rolling 30 days
AVAIL-02	Control plane — Institutional/Enterprise	99.95%	Rolling 30 days
AVAIL-03	Strategy execution workers	99.9%	Rolling 30 days
AVAIL-04	Exchange Connector Service	99.9%	Rolling 30 days
AVAIL-05	Notification dispatcher	99.5%	Rolling 30 days
AVAIL-06	BigQuery analytics pipeline	99%	Rolling 30 days

2.2 Recovery Objectives

ID	Metric	Target
RECO-01	Recovery Time Objective (RTO) — P1 incidents	< 15 minutes
RECO-02	Recovery Point Objective (RPO) — trade data	< 1 minute (Cloud Spanner synchronous replication)
RECO-03	Recovery Point Objective (RPO) — user config	< 5 minutes (Firestore multi-region)
RECO-04	Strategy worker pod restart	< 30s (GKE pod restart policy)
RECO-05	KILL_SWITCH activation to all workers paused	< 10s

2.3 Fault Tolerance

ID	Requirement	Implementation
FAULT-01	No single point of failure in control plane	GKE Autopilot multi-zone; min 3 replicas per service
FAULT-02	Exchange connector failover	If primary WebSocket fails, CCXT REST fallback within 5s
FAULT-03	Vertex AI fallback	If online prediction unavailable, static allocation model activates

ID	Requirement	Implementation
FAULT-04	Notification channel fallback	If WhatsApp fails, auto-fallback to SMS within 30s
FAULT-05	FX rate failover	If Open Exchange Rates unavailable, Wise API fallback within 5s
FAULT-06	Database write failure	Cloud Spanner write retried up to 3x with exponential backoff; alert on 3rd failure

3. Scalability

ID	Requirement	Target	Implementation
SCALE-01	Horizontal pod autoscaling	Scale from 0 to 1M concurrent workers	GKE Autopilot HPA on CPU + Pub/Sub queue depth metric
SCALE-02	Database scaling	Handle 50M trade writes/day	Cloud Spanner (linear horizontal scaling)
SCALE-03	User registration scaling	10M registered users	Firebase Auth (supports 100M+ natively)
SCALE-04	Zero-cost idle users	Registered users with inactive AI engine cost near \$0	GKE Autopilot scale-to-zero; no worker pods for inactive users
SCALE-05	Africa traffic elasticity	Handle 10x traffic spikes during naira devaluation events	GKE Autopilot auto-provision; Cloud CDN edge cache
SCALE-06	BigQuery analytics	Sub-10s P&L queries for any user across all trade history	Partitioned by user_id + date; clustered by strategy_type

4. Security

4.1 Data Security

ID	Requirement	Implementation
SEC-01	Credentials encrypted at rest	GCP Secret Manager AES-256 Google-managed encryption
SEC-02	Credentials encrypted in transit	TLS 1.3 on all connections; enforced by Cloud Load Balancer
SEC-03	Credentials never stored in application database	All API keys and OAuth tokens in Secret Manager only; Firestore stores only secret reference paths
SEC-04	Credentials never logged	Cloud Logging structured log schema excludes credential fields; log sanitization in application layer
SEC-05	Credentials never returned to frontend	API responses never include raw API keys; frontend only sees connection status
SEC-06	User data encrypted at rest	Firestore and Cloud Spanner: Google-managed encryption at rest by default

ID	Requirement	Implementation
SEC-07	PII isolation	User PII (email, phone, name) stored separately from trade data; different Firestore collections

4.2 Access Control

ID	Requirement	Implementation
SEC-08	Workload Identity for all GCP service auth	All pods use Workload Identity; zero static service account keys in any codebase
SEC-09	Namespace isolation	GKE NetworkPolicy prevents cross-namespace pod communication
SEC-10	Secret Manager access scoping	Each namespace service account can only read secrets under its own <code>/users/{uid}/</code> path
SEC-11	API authentication	Firebase Auth JWT required on every API endpoint; validated in Cloud Endpoints middleware
SEC-12	Admin access	Admin APIs protected by separate admin role claim in Firebase Auth custom tokens
SEC-13	VPC-SC perimeter (Institutional+)	Data plane cluster wrapped in VPC Service Controls perimeter

4.3 Application Security

ID	Requirement	Implementation
SEC-14	Input validation	All user inputs validated via Zod schema (TypeScript); server-side validation on all APIs
SEC-15	WAF protection	Cloud Armor: OWASP Top 10 ruleset active
SEC-16	Rate limiting	Per-IP rate limiting on Cloud Armor; per-user rate limiting in API middleware
SEC-17	SSRF protection	Exchange Connector Service only connects to whitelisted exchange domains
SEC-18	Dependency scanning	GitHub Dependabot + Snyk in CI/CD pipeline; no deployment with critical CVEs
SEC-19	Secrets scanning	GitHub secret scanning enabled; pre-commit hook blocks credential commits
SEC-20	Penetration testing	Annual third-party pentest; required before Institutional tier launch

5. Compliance

ID	Requirement	Jurisdiction	Timeline
COMP-01	FSCA CASP registration	South Africa	Before ZA public launch
COMP-02	SEC Nigeria Digital Asset registration	Nigeria	Before NG public launch
COMP-03	Kenya VASP Act registration	Kenya	Before KE public launch
COMP-04	GDPR compliance	EU	Before EU user acceptance

ID	Requirement	Jurisdiction	Timeline
COMP-05	POPIA compliance	South Africa	Before ZA public launch
COMP-06	NDPA compliance	Nigeria	Before NG public launch
COMP-07	FinCEN MSB legal opinion	USA	Before US marketing launch
COMP-08	SOC 2 Type II	Global (Institutional)	Month 12
COMP-09	SARS-compliant tax export	South Africa	Phase 3
COMP-10	Data retention policy	Global	Before launch: 12-month active, 7-year archived
COMP-11	Non-custodial legal opinion	Global	Before any launch
COMP-12	Nigeria P2P fiat leg legal review	Nigeria	Before automating fiat leg (Phase 5)

6. Observability

ID	Requirement	Implementation
OBS-01	Distributed tracing across all services	Cloud Trace via OpenTelemetry SDK; trace ID propagated through all service calls
OBS-02	Structured logging	All services emit JSON structured logs to Cloud Logging
OBS-03	Custom metrics	Trade execution latency, signal detection latency, execution quality score emitted as Cloud Monitoring custom metrics
OBS-04	SLO dashboards	Cloud Monitoring SLO tracking for all AVAIL-XX requirements
OBS-05	Error tracking	Sentry integrated in frontend (Next.js + React Native) and backend services
OBS-06	Product analytics	PostHog for onboarding funnel, feature adoption, retention cohorts
OBS-07	Alerting	PagerDuty: P1/P2 incidents (24/7 on-call); Slack: P3/P4
OBS-08	Audit logging	All admin actions + secret access attempts logged to Cloud Logging; immutable

7. Maintainability & Developer Experience

ID	Requirement	Implementation
MAINT-01	Infrastructure as Code	All GCP resources defined in Terraform; no manual console provisioning
MAINT-02	Kubernetes workload management	All GKE workloads defined in Helm charts
MAINT-03	CI/CD pipeline	GitHub Actions → Cloud Build → GKE rolling deploy; all merges to main trigger deploy

ID	Requirement	Implementation
MAINT-04	Zero-downtime deploys	GKE rolling updates with health checks; PodDisruptionBudget enforced
MAINT-05	Feature flags	LaunchDarkly (or GCP Remote Config) for gradual feature rollout
MAINT-06	Database migrations	Flyway for Cloud Spanner DDL migrations; backward-compatible only
MAINT-07	Dependency management	Dependabot for automated security patches; reviewed weekly
MAINT-08	Code quality	ESLint + Prettier (TypeScript); Ruff (Python); enforced in CI
MAINT-09	Test coverage	Minimum 80% unit test coverage; all strategy logic requires 95% coverage
MAINT-10	Documentation	All APIs documented in OpenAPI 3.0; all services have README with setup + runbook

8. Usability & Accessibility

ID	Requirement	Target
UX-01	Onboarding completion rate	> 85% of signups activate AI engine
UX-02	Exchange connection time	< 4 minutes from wizard start to validated connection
UX-03	Time to first trade	< 24 hours from AI engine activation
UX-04	Mobile-first design	All screens functional at 375px viewport; tested on Android Go (low-end)
UX-05	Low-bandwidth mode	Core dashboard functions on 3G (< 1MB page weight in low-bandwidth mode)
UX-06	Offline state	Cached P&L and trade history shown with "last updated" indicator when offline
UX-07	Accessibility	WCAG 2.1 AA compliance for web; VoiceOver/TalkBack support for mobile
UX-08	Localization	English (launch); French (Phase 2 — West Africa); Portuguese (Phase 3 — Mozambique/Angola)
UX-09	Zero jargon rule	No technical trading terms in user-facing copy without plain-language explanation

9. Africa-Specific Non-Functional Requirements

ID	Requirement	Target	Notes
AFR-01	GCP region for South Africa users	<code>africa-south1</code> (Johannesburg)	< 50ms latency to VALR/Luno APIs (benchmark required)
AFR-02	Nigeria/West Africa routing	<code>europa-west1</code> (Belgium) until GCP Lagos available	Monitor GCP Africa region expansion roadmap
AFR-03	WhatsApp primary notification	100% of NG/KE/GH/TZ/UG/RW users	Fallback to SMS within 30s if WhatsApp delivery fails
AFR-04	Mobile money payment success rate	> 95% successful payment confirmations	Monitor Flutterwave + M-Pesa webhook success rates
AFR-05	NGN exchange rate freshness	P2P rate updated every 5 minutes	Open Exchange Rates + Binance P2P API combined
AFR-06	Low-end Android performance	App usable on Android 9+ with 2GB RAM	React Native Hermes engine; image lazy loading; minimal animations
AFR-07	Africa P2P data availability	Historical P2P premium data from Day 1	Cloud Function collector starts before any other Africa feature
AFR-08	SARS tax export availability	Available in dashboard by Phase 3	Required for South Africa regulatory compliance

10. Disaster Recovery

ID	Scenario	Response Plan	RTO	RPO
DR-01	GCP region outage (<code>africa-south1</code>)	Traffic fails over to <code>europa-west4</code> ; Africa users experience higher latency but no data loss	< 5 min	< 1 min
DR-02	Cloud Spanner regional outage	Multi-region config (<code>nam-eur-asia1</code>) provides automatic failover	< 1 min	0 (synchronous replication)
DR-03	Vertex AI endpoint unavailable	Static allocation model activates; no new Vertex AI predictions until restored	< 1 min	N/A
DR-04	Exchange API outage (Binance)	Binance strategies paused; other exchanges continue; users notified via WhatsApp/Telegram	< 5 min	N/A
DR-05	Stripe outage	Flutterwave backup for Africa; new subscriptions queued; existing subscriptions unaffected	< 30 min	N/A
DR-06	Complete platform outage	KILL_SWITCH triggered; all strategies paused; status page updated; all users notified via SMS	< 15 min	< 1 min
DR-07	Security breach (unauthorized secret access)	Affected secrets rotated immediately; impacted users notified; legal counsel engaged	< 30 min	N/A

11. Cost Management

ID	Requirement	Implementation
COST-01	GKE scale-to-zero for inactive users	Worker pods terminated when AI engine inactive; cost = \$0 for idle users
COST-02	BigQuery cost control	Partitioned tables; query cost estimates enforced before dashboard queries run
COST-03	Vertex AI cost monitoring	Budget alerts at 80% of monthly Vertex AI budget; auto-scaling caps per tier
COST-04	Notification cost monitoring	WhatsApp conversation cost tracked per user per day; alert if cost/user exceeds \$0.05/day
COST-05	GCP budget alerts	Budget alerts at 50%, 80%, 100% of monthly GCP budget; Slack notification
COST-06	Cost per user tracking	Monthly cost per active user tracked per tier; alert if gross margin drops below 70% for any tier

NFRD v1.0 — June 4, 2026. All requirements subject to review and revision by Tech Lead prior to Phase 1 kickoff.